DG INTERN HUB

SOC ANALYST INTERNSHIP

WEEK 2 —NOTES

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Submission Date: 15 October 2026

Common Log Formats

Learning Objective

To understand common log formats used in cybersecurity and learn how SOC analysts use different types of logs during investigations.

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1. What are Logs?

Logs are records of events or activities that happen on a system, application, server or network.

For example, a log can record:

- User login
- Failed login attempt
- Application error
- Network connection
- File access
- System changes
- Security events

SOC analysts use logs to investigate suspicious activity and understand what happened on a system.

Different systems can store logs in different formats.

The four formats I learned about are:

- Windows Event Log
- Syslog
- JSON
- CSV

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2. Windows Event Log

Windows Event Logs are logs generated by Windows systems.

They contain information about different events happening on a Windows computer.

Common logs include:

- Security
- System
- Application

For example, a failed Windows login can generate:

Event ID: 4625

A successful login can generate:

Event ID: 4624

These logs are very useful for SOC investigations because they can contain information such as:

- Username
- Login time
- Logon type
- Source IP
- Event ID
- Computer name

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3. Syslog

Syslog is a common logging standard used by Linux systems, network devices, firewalls and other systems.

Syslog messages can contain information such as:

- Timestamp
- Hostname
- Application/service
- Message
- Severity

For example:

Oct 8 12:20:10 ubuntu sshd[1234]: Failed password for user

As a SOC analyst, I can use syslog data to investigate things like failed SSH logins, system events and network device activity.

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4. JSON

JSON (JavaScript Object Notation) stores information using key-value pairs.

It is commonly used by:

- APIs
- Cloud services
- Security tools
- Applications
- Modern logging platforms

Example:

```
{
  "user": "admin",
  "action": "login",
  "status": "failed",
  "source_ip": "192.168.1.10"
}
```

JSON is useful because the information is structured and individual fields can be easily processed by tools such as SIEM platforms.

For example, I can filter logs based on:

source_ip
user
action
status

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5. CSV

CSV (Comma-Separated Values) stores information in rows and columns.

Example:

Time,User,Action,Source_IP
12:20,admin,login,192.168.1.10
12:25,user1,login,192.168.1.20

CSV files are easy to open in applications such as Excel or other spreadsheet tools.

They can be useful when analysing exported security logs or datasets.

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6. Comparison of Common Log Formats

Format	Structure	Common Use Case	Example
Windows Event Log	Structured Windows event records	Windows security and system monitoring	Event ID 4625 — Failed login
Syslog	Timestamp + host + service + message	Linux, firewalls, routers and network devices	Failed SSH login

JSON	*Key-value pairs*	*Applications, APIs, cloud and security tools*	*"status": "failed"*
CSV	*Rows and columns separated by commas*	*Exported logs and data analysis*	*admin,login,192.168.1.10*

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7. Why Log Formats Matter in a SOC

A SOC analyst may receive logs from many different sources.

For example:

Windows → Event Logs

Linux → Syslog

Cloud/API → JSON

Exported dataset → CSV

The analyst needs to understand the format so that they can find the important information.

For example, when investigating a failed login:

- In Windows logs, I might look for **Event ID 4625**.
- In Linux logs, I might search for **failed authentication messages**.
- In JSON logs, I might filter the `status` field.
- In CSV, I might filter the relevant column.

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8. Logs in SIEM

SIEM tools collect logs from different sources and allow analysts to search and analyse them in one place.

For example, Splunk can receive logs from:

- Windows
- Linux
- Firewalls
- Applications
- Network devices

Once the logs are collected, an analyst can search for patterns and suspicious activity.

For example:

Failed login attempts

↓

Check source IP

↓

Check username

↓

Check time

↓

Check number of attempts

↓

Investigate if suspicious

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9. My Understanding

I learned that logs are basically records of what is happening inside systems and networks.

The format can be different depending on where the logs come from.

For example, Windows uses Event Logs, Linux commonly uses syslog, applications may use JSON, and exported datasets can use CSV.

As a SOC analyst, I need to understand these formats because the useful information can be stored differently in each one.

I also understood why SIEM tools are useful because they can bring logs from different sources together and make it easier to search and investigate them.

10. Practical SOC Investigation Example

Suppose I receive an alert that there are multiple failed login attempts on a Windows system.

The first thing I would check is the Windows Security Event Log.

For example:

Event ID: 4625
Account: admin
Source IP: 192.168.1.50
Logon Type: 3
Time: 12:30 PM

I would then investigate:

1. How many failed login attempts happened?
2. Which account was targeted?
3. What source IP was generating the attempts?
4. Were the attempts happening within a short period?
5. Was there a successful login after the failed attempts?
6. Is the source IP known or suspicious?
7. Were there any other suspicious events from the same system?

If I have the logs in Splunk, I could search for the failed login events and group them by account or source IP.

For example:

index=main EventCode=4625

| stats count by Account_Name, Source_Network_Address
| sort - count

This can help me identify accounts or IP addresses with a high number of failed login attempts.

If I see something like:

admin 25 failed attempts
user1 2 failed attempts
user2 1 failed attempt

I would investigate the `admin` account further instead of assuming it is automatically an attack.

The same investigation can also involve different log sources. For example, I could compare Windows authentication logs with firewall or network logs to see whether the source IP was also making other suspicious connections.

This shows why understanding log formats is important. The format tells me how the information is structured, but the actual investigation requires me to connect the information from different logs.

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